

The Printed Circuit Board (PCB) Layout --- A Tutorial

This project demonstrates how we produce an electronic project using one particular PCB CAD package. Our business is about producing finished custom electronics, and the PCB design process is an important part of that. This page might be of interest to new designers wanting to learn principles, or to potential clients wanting to understand our processes more thoroughly.



There is a nasty little "bug" in the circuit board described here - one of those horrible things that you **can't do much about:**

Windows doesn't control the serial port handshaking lines very well. Actually, compared with DOS, it **is very poor in its timing.** Any **RTS controlled** RS485 converter (and, like ours, that is how most of them work!) will only have mediocre performance under windows - and with some RS485 devices at higher baud rates it may well fail to operate. Our advice is *test it completely* before you commit to making any quantity...having said that this page is still useful for teaching principles of CAD, and for those who want an RS485 converter controlled by their own specially written drivers -

If you want to use this design, please [contact us](#) for schematics

The project we have chosen as our example is an RS232 to RS485 converter. This is a often needed device which can demonstrate a lot of principles of circuit design, and hopefully the finished project will be really useful to a lot of people, just as it stands.

We start with a specification - to do things "by the book" - we should define our goals and set limits.

RS232 to RS485 Converter specification

- DB9 RS232 connector compatible with PC
- TxD and RxD translated from RS232 to RS485
- Direction of transmission controlled by PC RTS line
- Handshake loop the PC connection so it works with all software
- RS485 signals output on 6 way modular jack
- Indicator LED(s) to show communications traffic
- Power supplied by external unregulated 9v plugpack
- ... and a few extra features, not essential, but desirable
 - RS485 Termination resistor, jumper selectable
 - Pullup/Pulldown resistors on RS485 to establish line-idle condition

As the design proceeded we added a couple of extra features over and above the specification, for the sake of completeness they were:

- RS485 Transmission not just controlled by RTS, but also switched to 'Transmit' whenever the RS232 TxD line goes to space (active). While special software running on the PC should use the RTS line to control RS485 Tx, passive mark pulling and RS485 Tx activated by RS232 space allows a quick and easy test of RS485 serial COMs with existing software.
- Take RI and ground through to spare pins on the RS485 jack

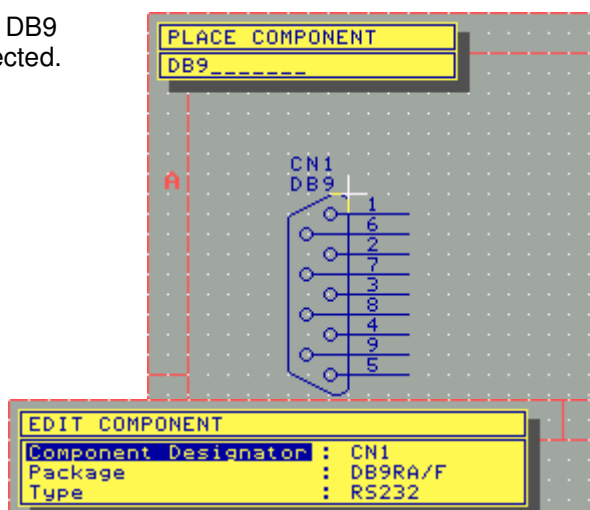
By using a Schematic CAD Package to create the circuit diagram, then generating a Netlist to use as the input to a PCB CAD package, the designer has the benefit of being able to run a Design Rule Check to verify that the final PCB matches the original circuit.

Most engineers choose to start the circuit design on paper. Some designers, especially when starting out and attempting their first simpler PCBs, go straight from paper to PCB CAD with no netlist. This can work well as long as the PCB design is checked rigorously.



We create a new schematic diagram, allocating a name to the project, a version number to the schematic, and a drawing number for the printed circuit board. We also put on our copyright/logo, and a symbol that we use to record the PCB drawing number

We place components on the schematic, for instance the DB9 connector through which the RS232 signals will be connected.



As we place the components, we edit the package details (and silkscreen legend 'Type') to match what we want on the Printed Circuit Board.

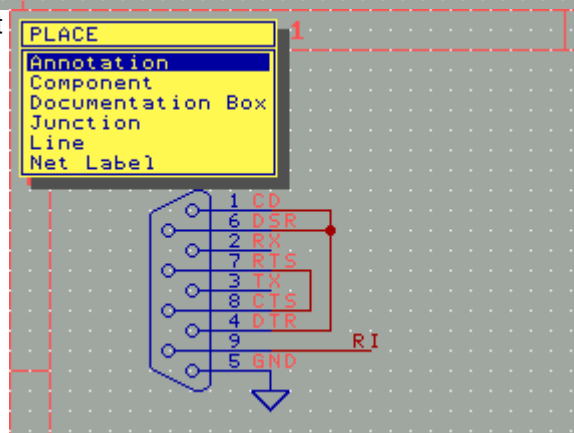
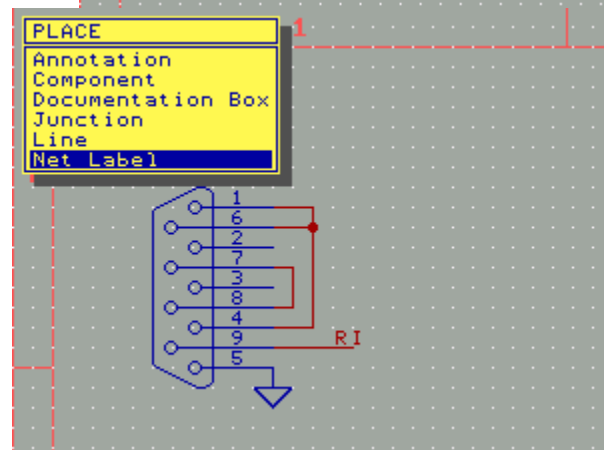
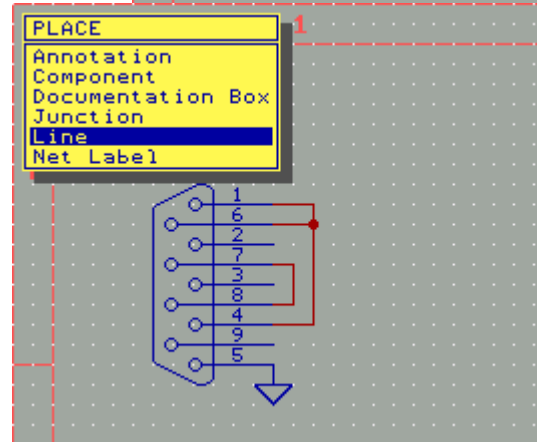
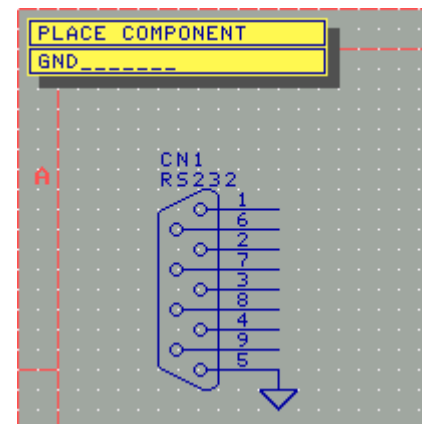
Items such as ground symbols, test points and jumpers are placed as different types of components. The copyright/logo and PCB drawing number symbol (placed previously) were also different types of components

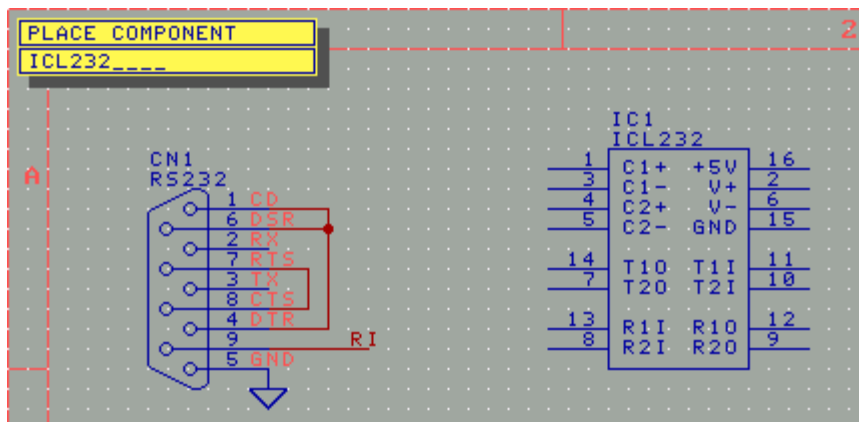
Lines are placed to connect components, taking care to start the line exactly on the end of the component pin.

Lines can be inter-connected by placing junctions.

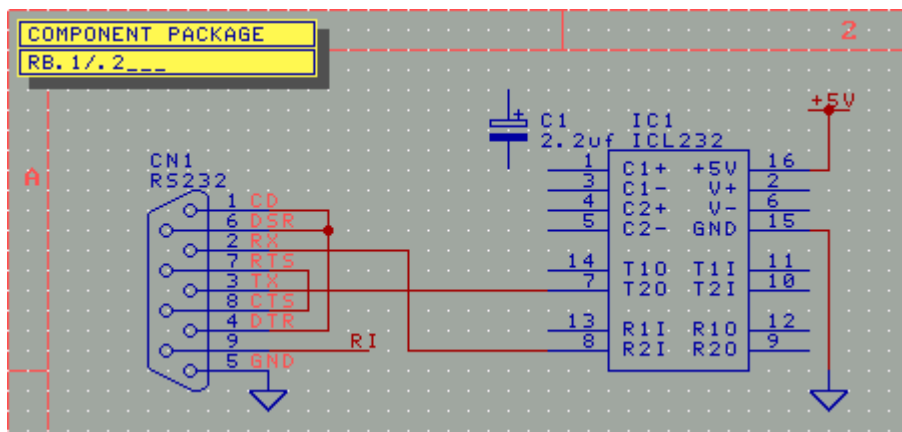
Signals that are not easily connected by lines may be connected by using a net label. A net label (e.g. '+5V'; or 'RI' - ring indicator signal in the schematic below) will connect all points of that name, even across multiple sheets if there is more than one sheet in the schematic.

Annotation can be placed anywhere on a schematic, and it often helps to understand what signals are on what pins. Depending on your CAD package, it may be easy to confuse Annotation and Net labels - just remember that annotation is a note for the readers benefit - it does not connect lines the way net labels do.

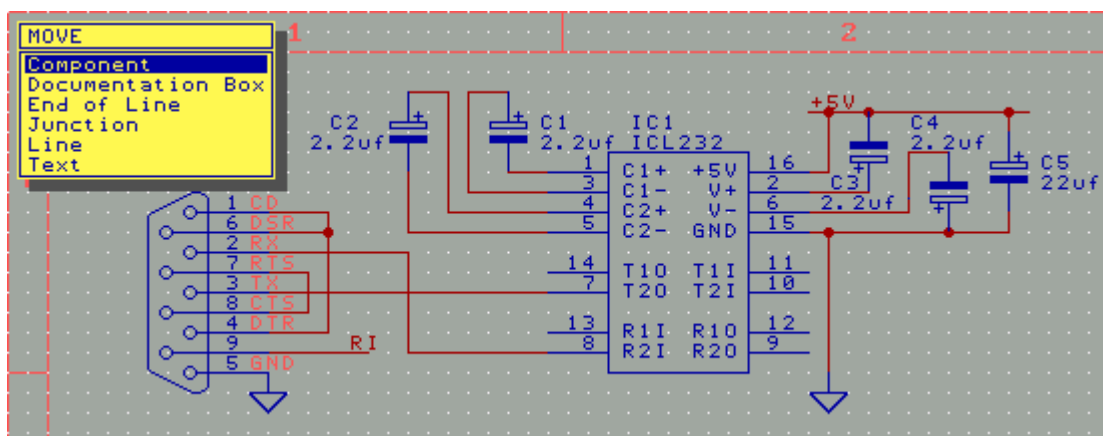




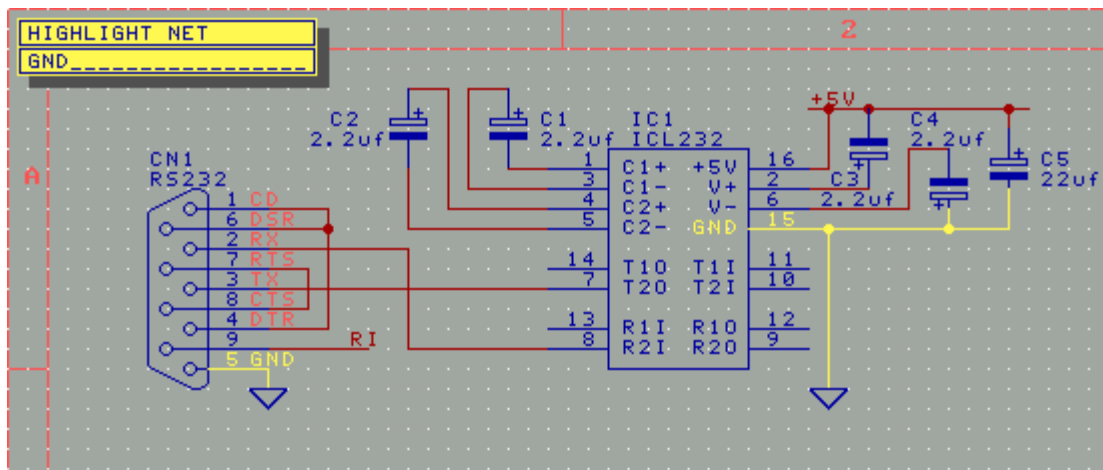
Reasonably extensive component libraries are available for most CAD packages. In any case, check the components pinout against current data sheets. Where components are not in the library, most CAD packages allow you to define your own components.



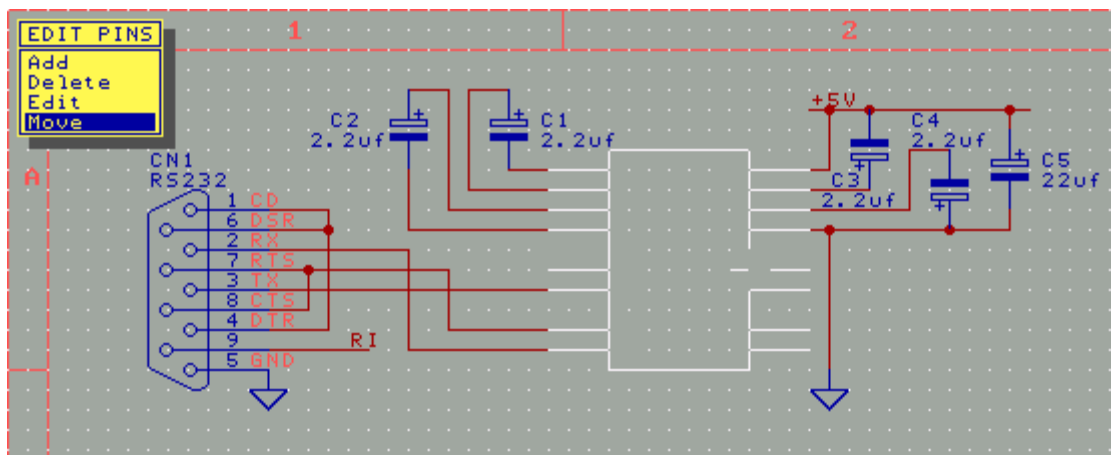
Some components have a choice of packages available, and it is important to consider the selection carefully. Normally the component package type is selected at the schematic design stage, and then carried through to the printed circuit board.



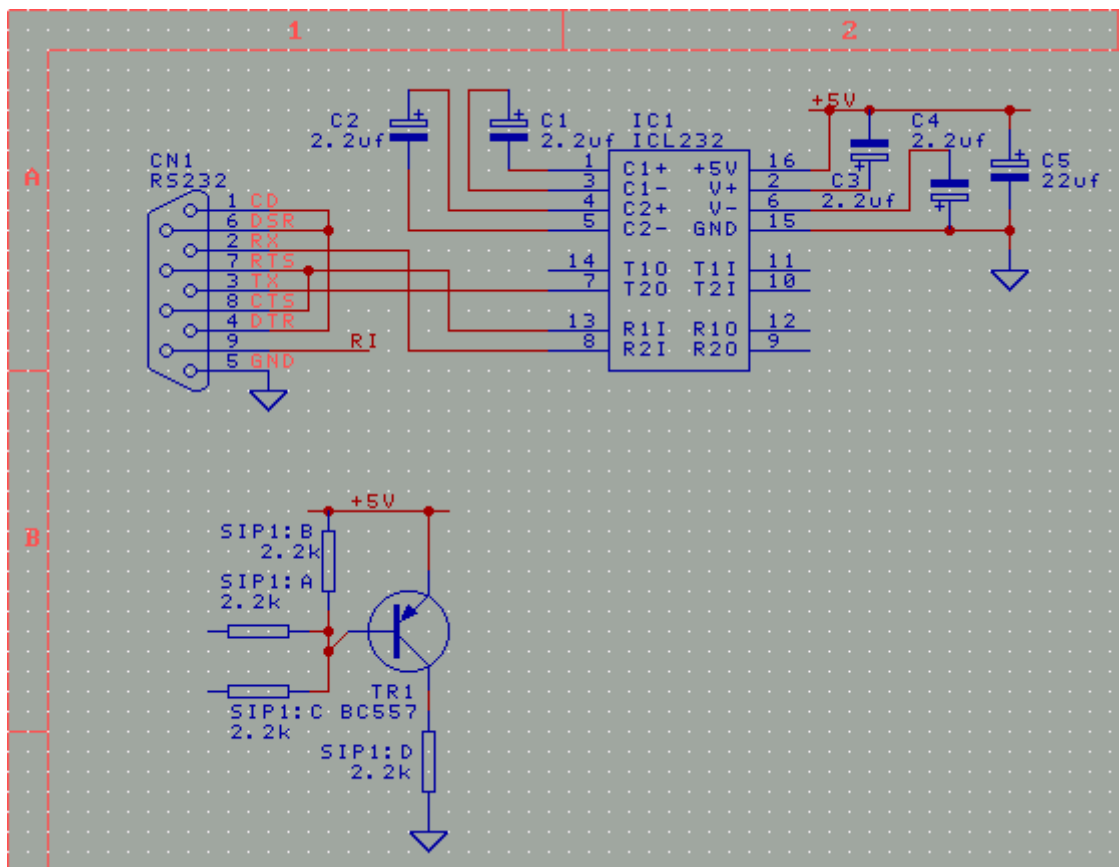
As the schematic progresses, text and components are ofcourse moved around to get a good circuit layout.



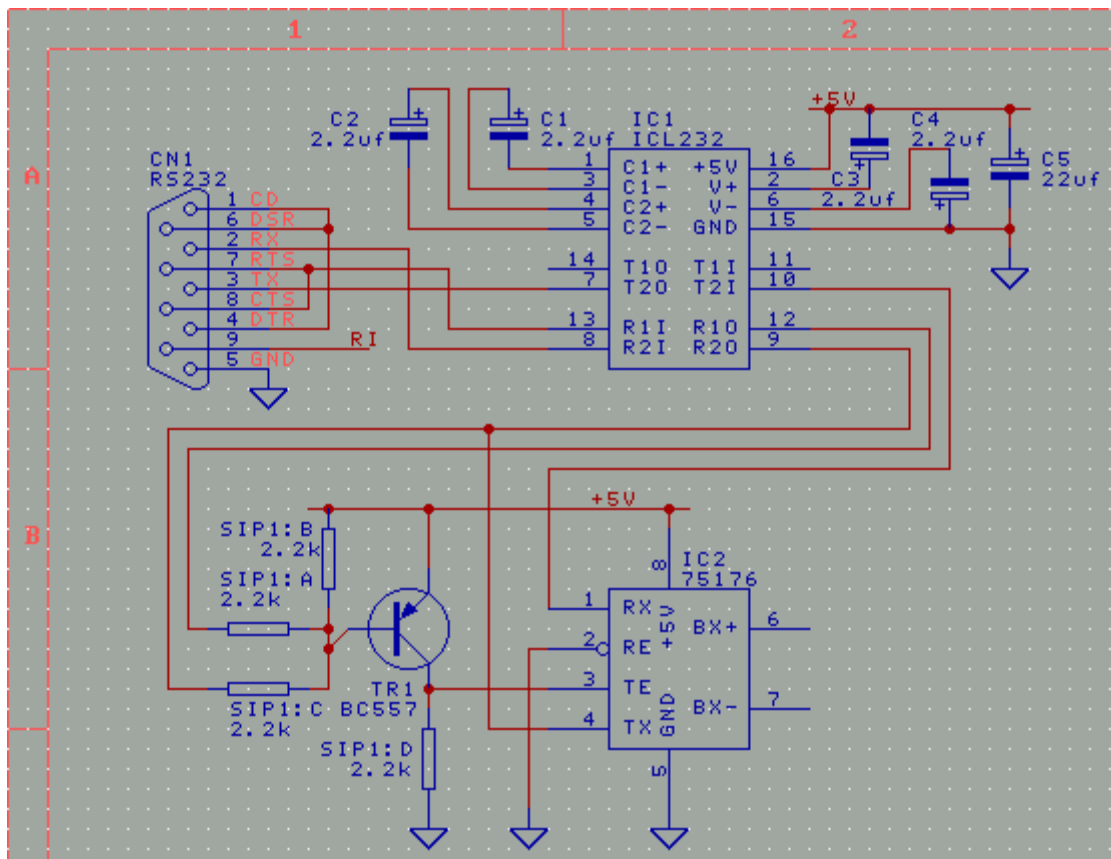
The net connection functions of the circuit diagram can be checked by the Highlight Net function (may vary from package to package)



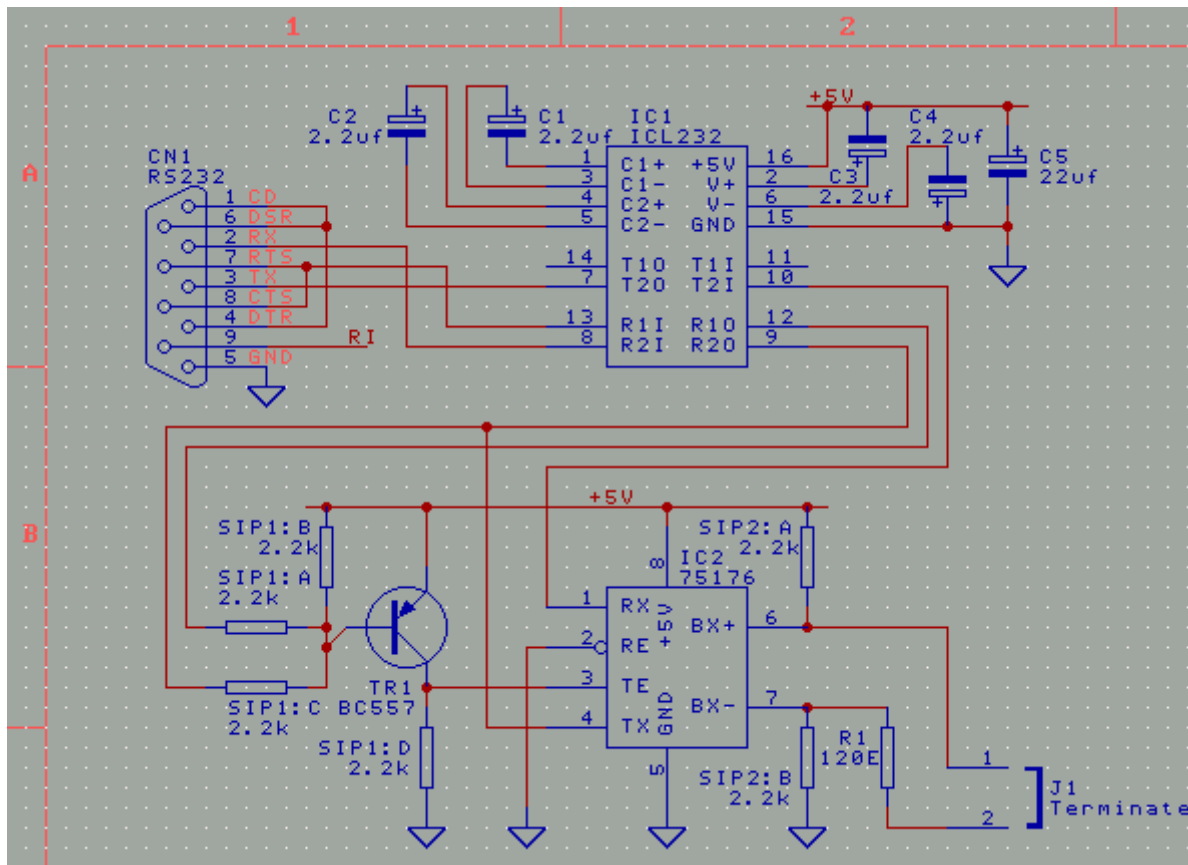
Existing library components can be edited (if required) to better fit the circuit diagram - for instance by moving pins



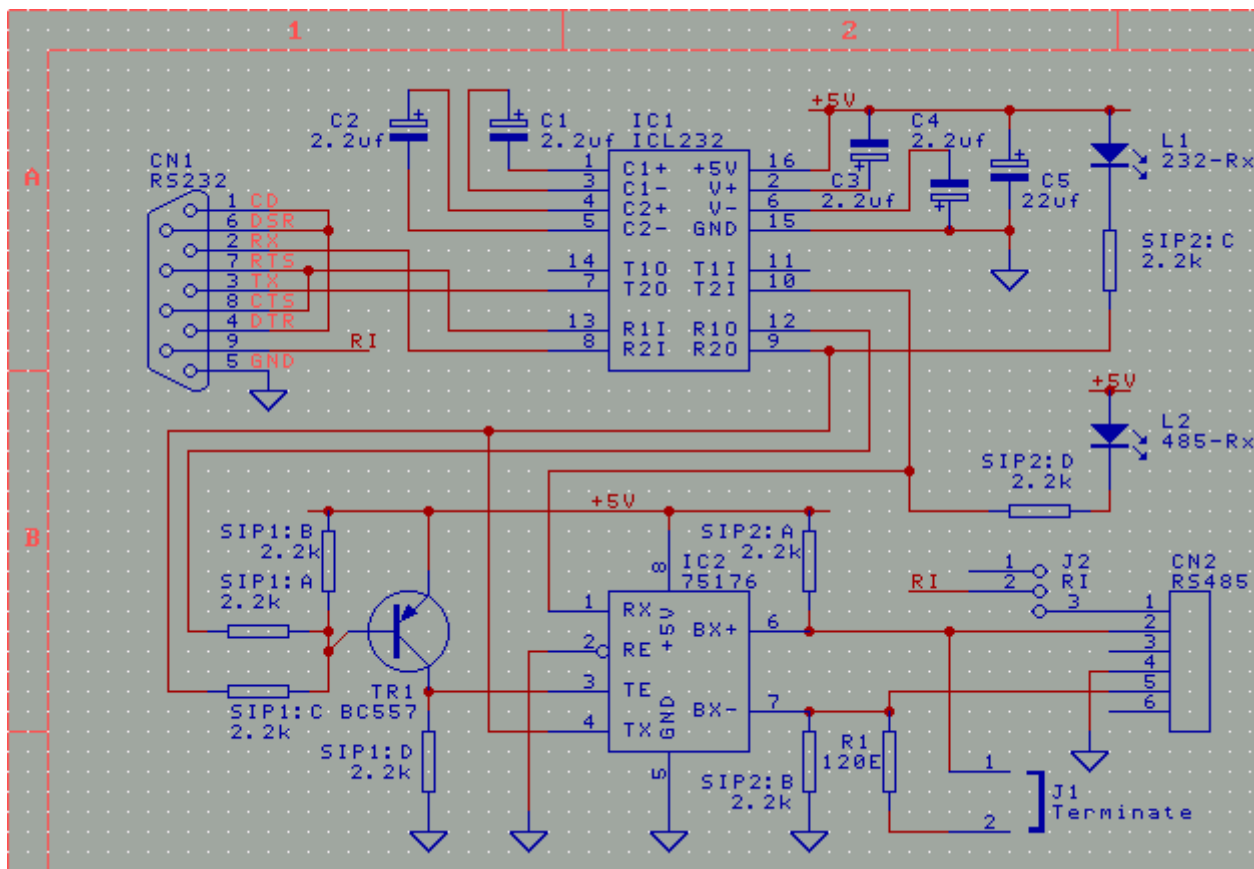
Some components have multiple parts - for instance individual resistors within a Single-Inline-Package resistor network, or individual gates within a logic chip. These may be indicated by a notation such as SIP1:B



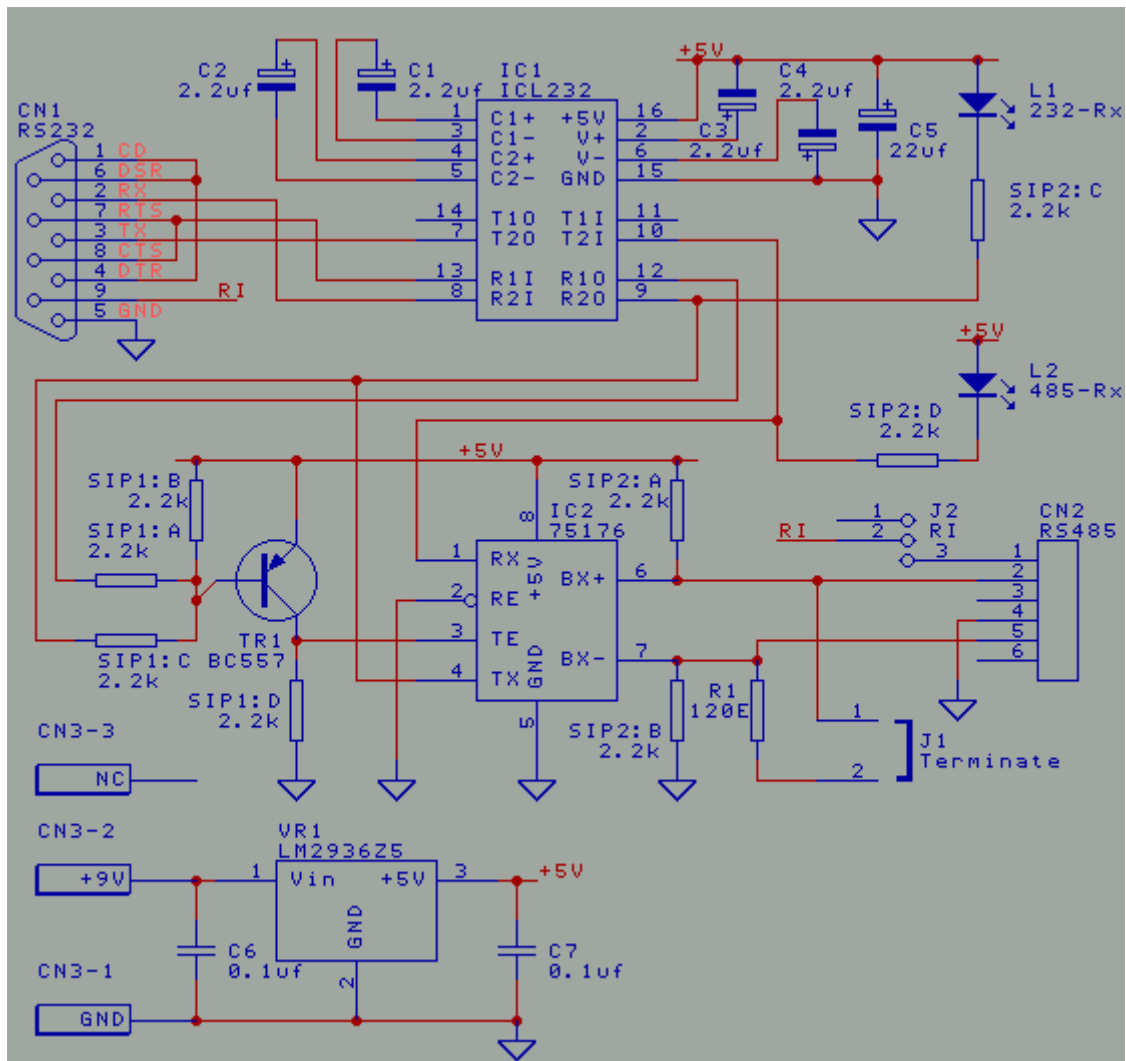
A good schematic can show connectivity in addition to its main task of showing the functionality of the circuit. This connectivity is part of the PCB layout task, but it is being thought through at the schematic design stage.



Resistor values are usually shown without the 'ohms' symbol - '2.2k' is a 2.2 kilo-ohm resistor, 120E is a 120 ohm resistor (other engineers may show this as 120R or just 120)



See 232cnv15.gif Schematics can be made more useful by descriptive labelling of connectors, option jumpers and indicators. This labelling will then carry through to the Printed Circuit Board



A power supply, power connector or batteries will be necessary for any project. The power supply will also need decoupling capacitors.

When the schematic is complete, the project should be transferred to a Netlist for importing into the PCB CAD package. The details of this step are very dependent on the particular CAD packages you are using.

```
-----
C:\SCRESHOT>dir
```

```
Volume in drive C has no label
Volume Serial Number is 01B2-0295
Directory of C:\SCRESHOT
```

```

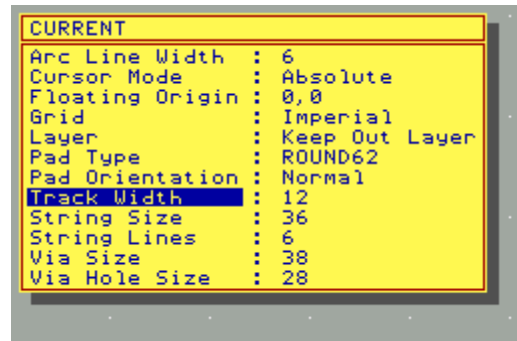
.           [DIR]          12-10-98  12:16p  .
..          [DIR]          12-10-98  12:16p  ..
NOTES      TXT             4,244  12-11-98  2:44p  notes.txt
232CNV     NET             1,765  12-11-98  2:53p  232CNV.NET
232CNV     S01            12,304  12-11-98  2:45p  232CNV.S01
ARCHIVE    [DIR]          12-10-98  3:47p  ARCHIVE
232CNV     WIR             2,738  12-11-98  2:53p  232CNV.WIR
232CNV     REP             582   12-11-98  2:53p  232CNV.REP
232CNV     BOM             1,117  12-11-98  2:53p  232CNV.BOM
6 file(s)                22,750 bytes
3 dir(s)                  4,172.28 MB free
```

```
C:\SCRESHOT>c:\schmatic\post 232cnv 232cnv
-----
```

Before starting the Printed Circuit Board layout, the settings for the CAD package should be initialized. This is dependent on the type of circuit board you wish to produce, and the details are dependent on the type of CAD package that you use.

In the example here we are showing the 'Current' settings. You will note our example is mostly going to use 12 thou tracks. A simpler loosely laid out circuit board might be laid out with 22 thou tracks. A More compact, tighter circuit board layout would normally use 8 thou tracks.

You will note we are using vias with a hole diameter of 0.7mm (28 thou) and a quite small pad size of 38thou. Other designers might tend to use a larger pad size for the via. For tight layouts, a small pad and hole would be used - typically a 16 thou hole (0.4mm) and a 28 thou pad

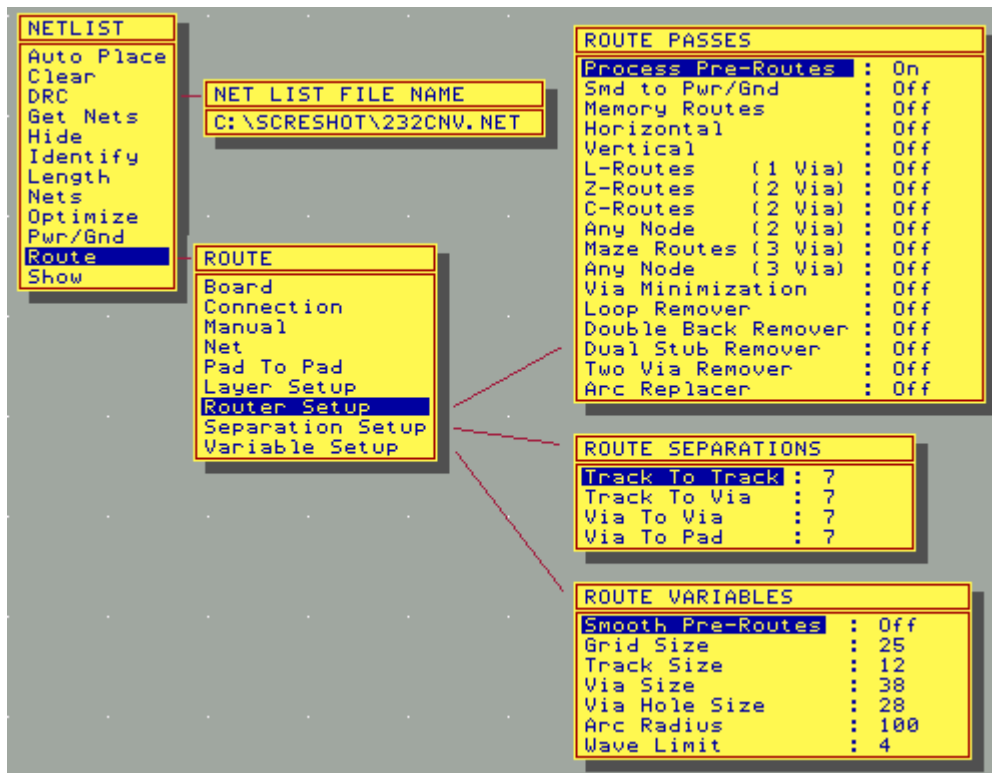


If you are using the CAD package we have shown in our example, make sure all your string sizes are in multiples of 12 thou - e.g., 36, 48 or 60 thou - as there is a bug in the package (!?) that rounds to this value

An imperial grid at 25 thou is usually the best coordinate system for standard printed circuit boards. As the density increases, designers usually drop to a 5 thou grid.

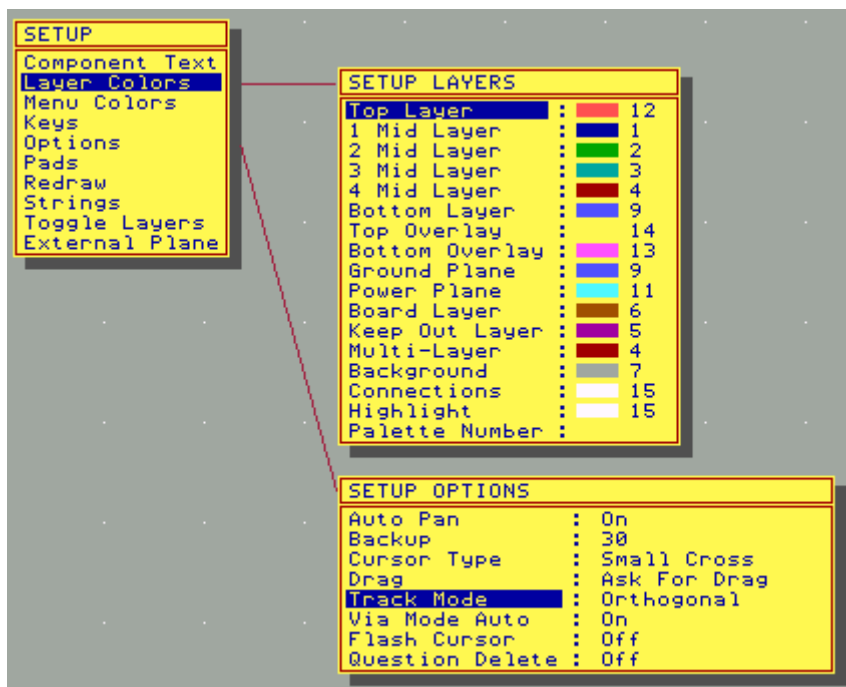


Again, Depending on your CAD package, there will be setup required in other areas involved with Netlists and Routing. To some degree, these options depend on how you will use the CAD package. In the particular package we are showing as an example we would not use the autorouter for more than checking completion progress - hence the options set.

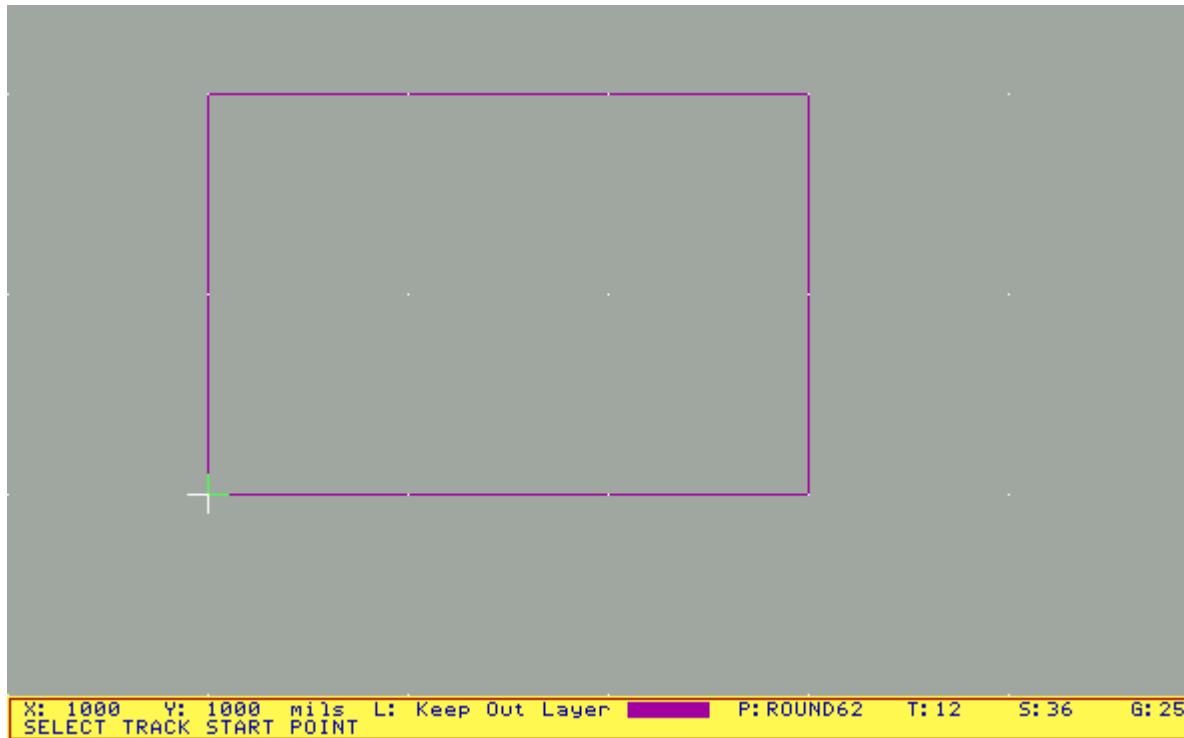


You will also have to ensure the components to be used in the PCB are available in the CAD parts Library.

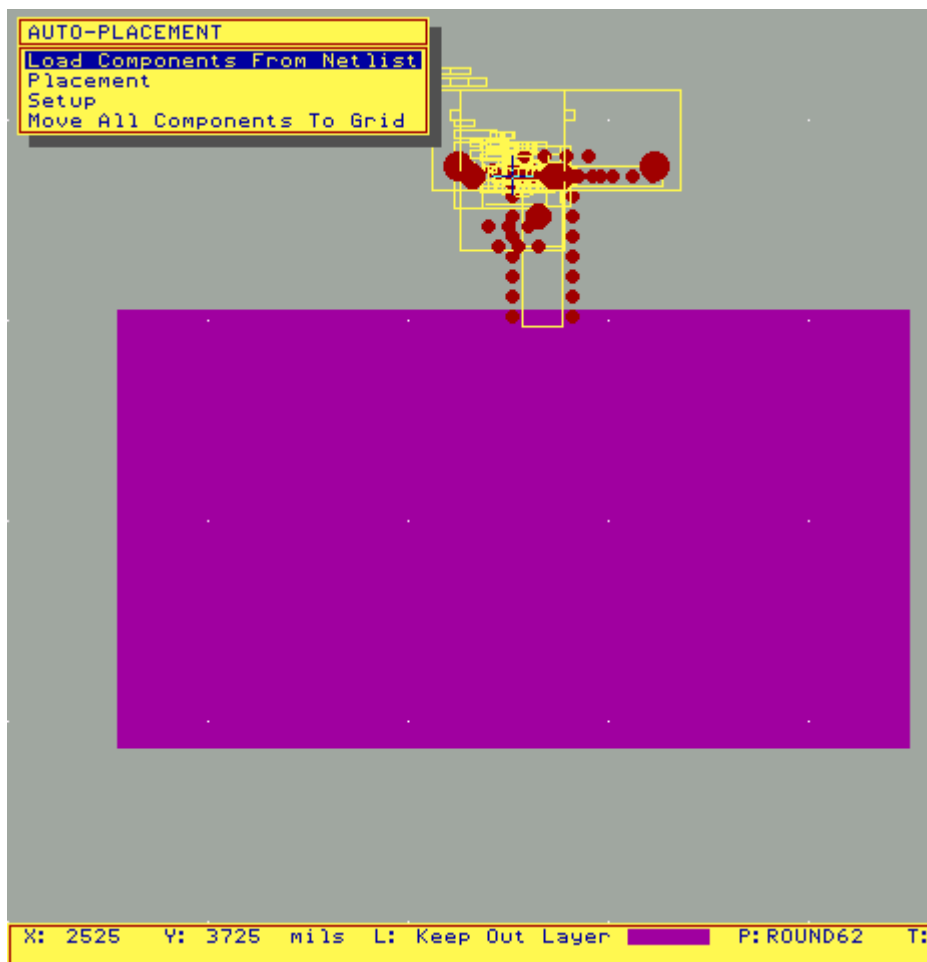
If you use the CAD package shown in the example, you should be aware that there is a limit on how many components may be in a library. You may either produce a library for each individual project (which you may find quite tedious) or you may load each library in turn, and run the 'N-A-L' keyboard command sequence (Netlist-Autoplace-LoadfromNet) on each library to load the components used in that library. This also allows you the opportunity to have more libraries, you can always sign up for ours: [AirBorn Electronics Autotrax Library](http://www.airborn.com.au/layout/pcbtrax)



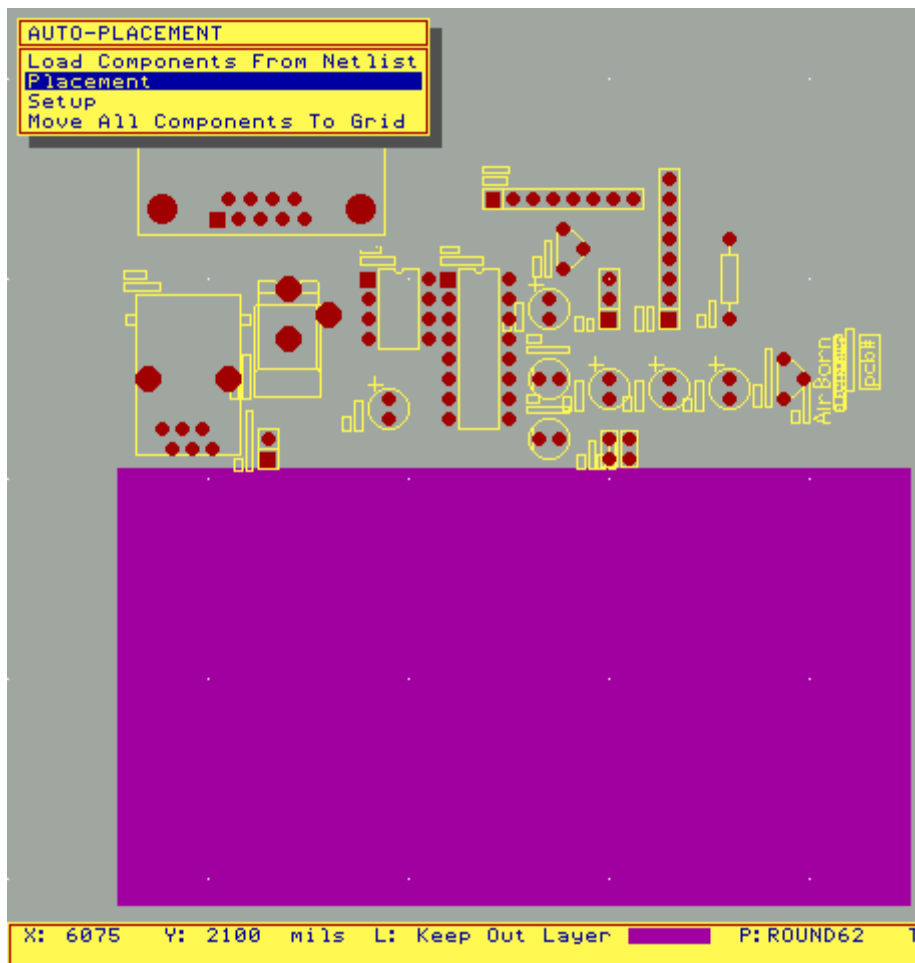
There will also be options to setup personal preferences such as colours.



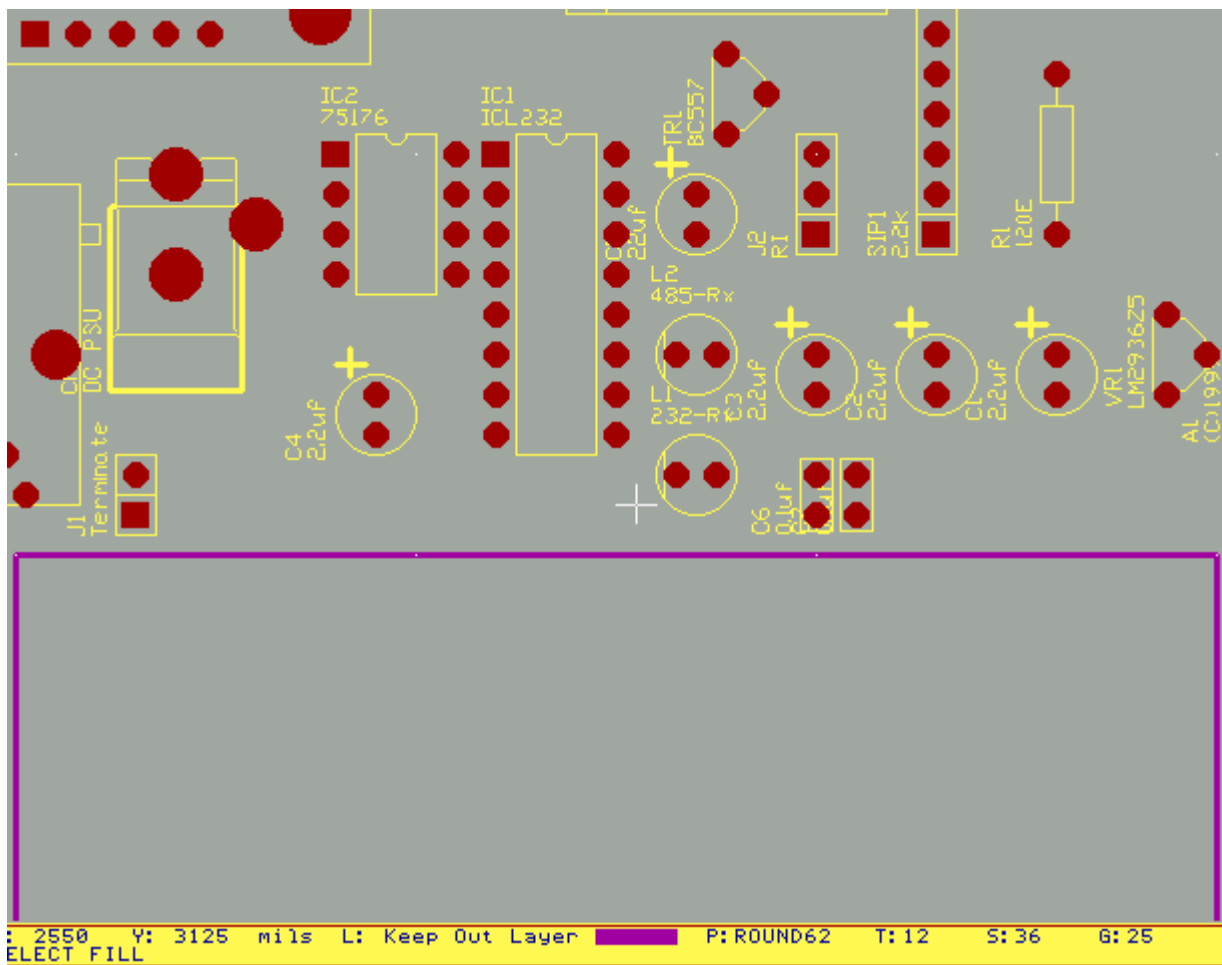
Loading the netlist will be very dependent on the particular CAD package you are using. In our example CAD package, the first step is to place a basic border on the keep out layer. In our example we took a guess at 2 inches by 3 inches - we can change it later. Remember to save your work frequently as you proceed through the next stages if you are following our example.



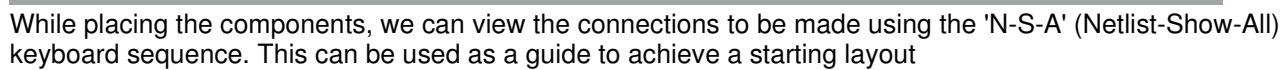
For our example we place a 'Fill' on the keepout layer, covering the whole of the circuit board. Then we do the 'N-A-L' keyboard sequence (Netlist-Autoplace-LoadfromNet) to load the components. The components are pulled from the library and piled in a heap at the cursor position - we put them above the circuit board.



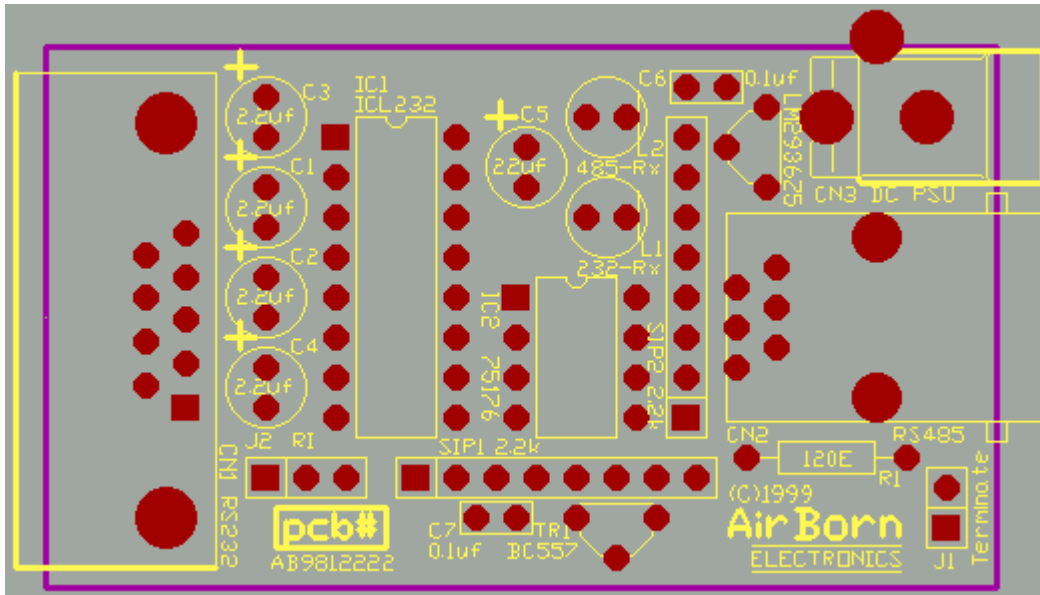
After executing the 'N-A-L' sequence to load the parts from the library on to the board, we proceed to a 'N-A-P' (Netlist-Autoplace-Placement) to spread the components out so we can see them easily.



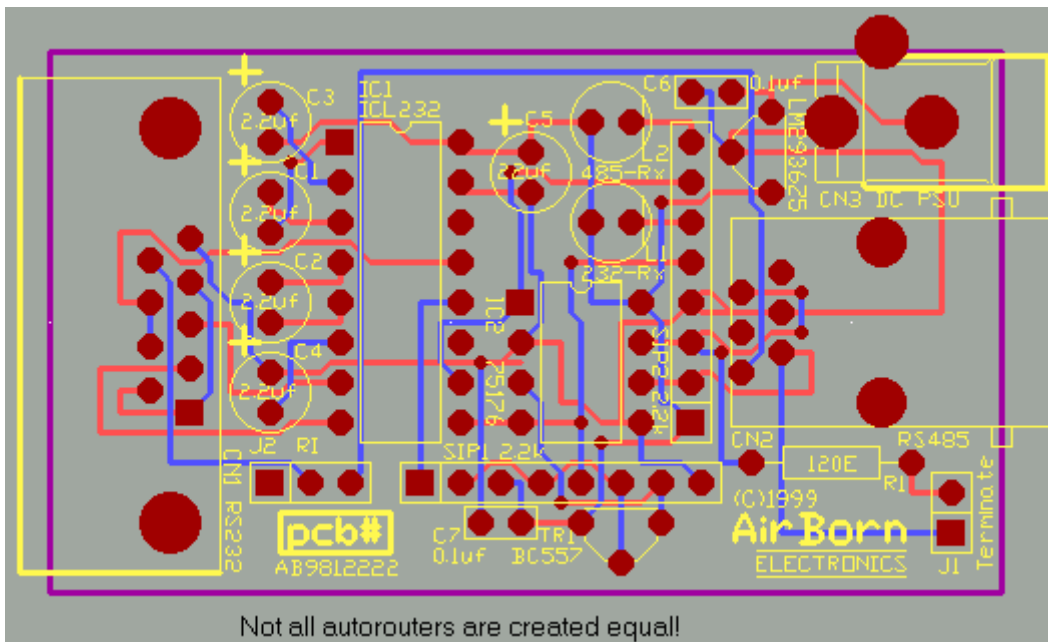
After executing the 'N-A-P' sequence we delete the fill that we had covering the circuit board. We now have all the components read in, and we can begin to place them. If there were any problems we would have encountered an error message, and the details would have been written to a .REP (report) file



After a good layout has been achieved, usually the border will be moved in to get a smaller board size. The labels will be moved around so that a good legend can be printed on the PCB

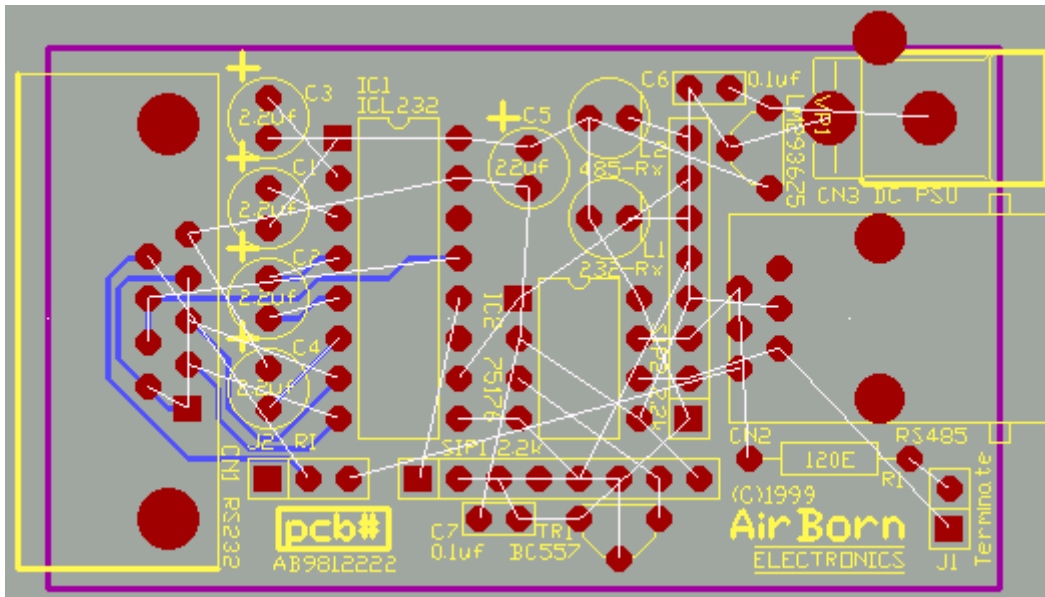


Designers will frequently autoroute part of a Printed Circuit Board layout. Even the best autorouters require a great deal of supervision and certainly some amount of cleaning up afterwards. And then there are some autorouters that are just plain shoddy and shouldn't be used:

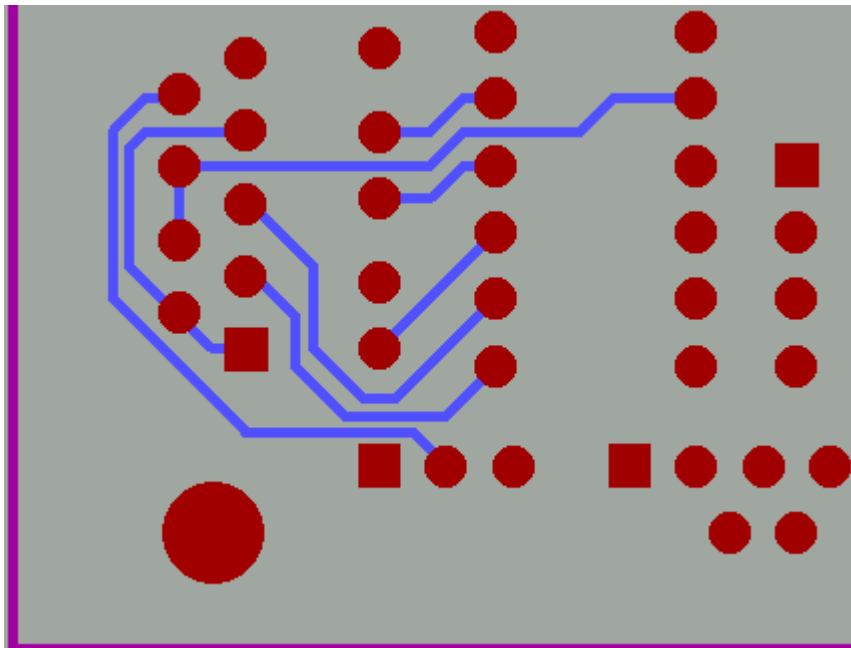


There are some areas of any circuit design that really should not be autorouted. There are some designs where none of the circuitry, at all, should be autorouted. The attraction of autorouters is that they save design time, it is very tempting to hit "Go" and to think you can have your board finished shortly afterwards, with no effort at all.

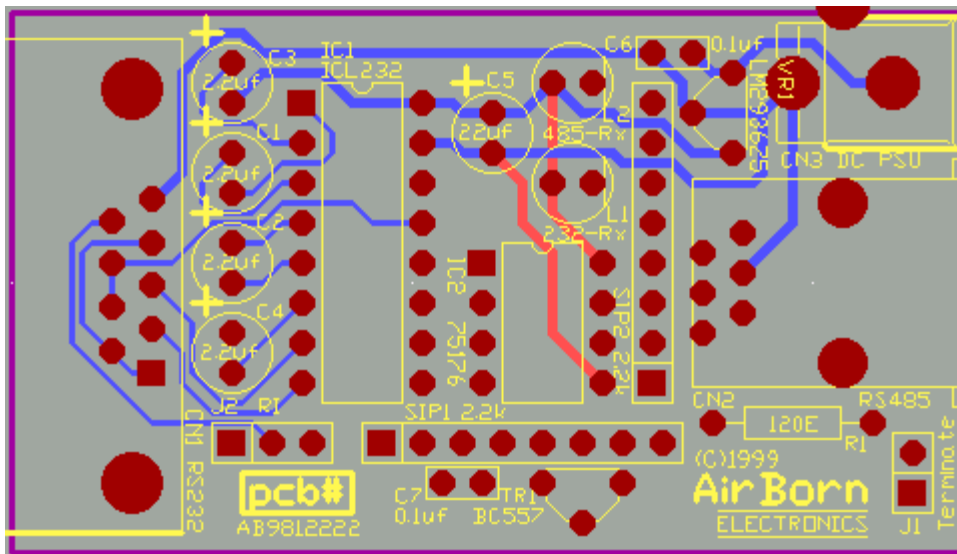
Beware any company that tells you their router is atleast as good as manual placement by a design engineer - remember those same software vendors also force you to waive statutory rights and most warranties before you open the shrinkwrap!



Manual routing uses the netlist as a guide. Manual routing can take intelligent advantage of contours, weighing complex decisions about track currents, noise pickup and grounding loops to achieve the best pattern. Tracks usually run on both sides of the printed circuit board (tracks on the solder side of the board are seen here in blue) however, some boards are simpler single sided, and other boards are more complex multilayer boards - most with four layers, but some with six or more layers

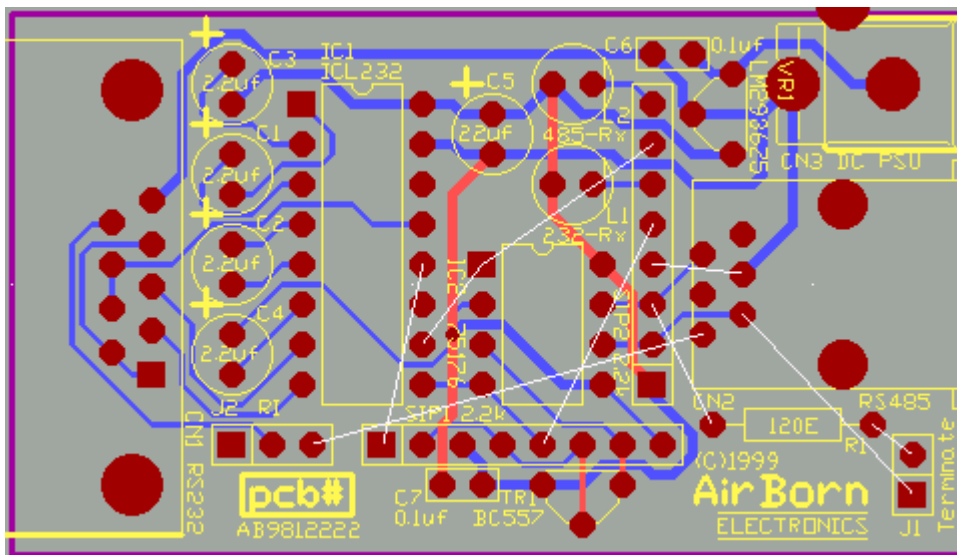


Layers can be switched on and off in the CAD package so that it is easier to see the tracks that are being routed. As we mentioned before, our example is mostly going to use 12 thou tracks - most of our PCB is spaced '12 by 12' ie 12 thou tracks and 12 thou clearance between tracks/tracks and tracks/pads. Another way of referring to the tolerance is 12 thou tracks on a 25 thou spacing (the grid size)

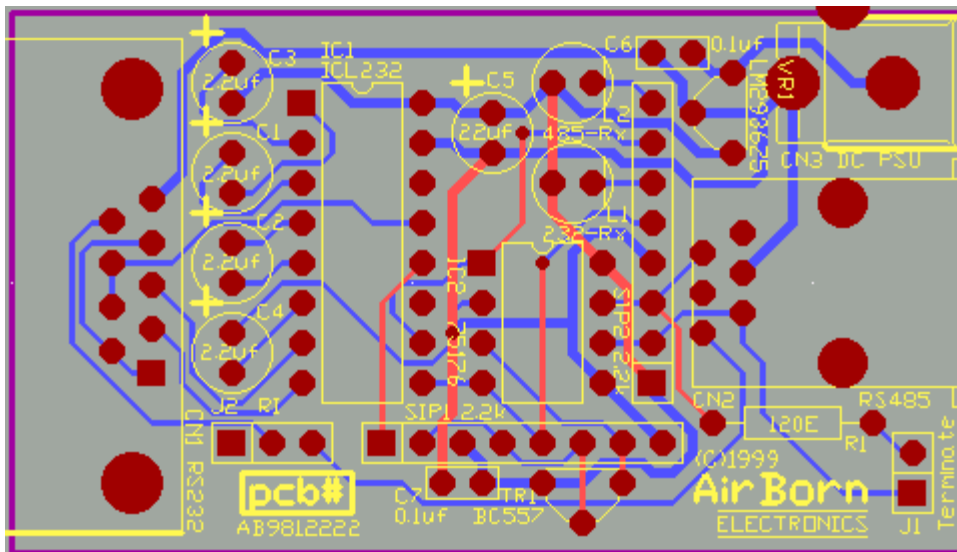


Heavier tracks are usually used for Power and Ground connections, in order to allow for the heavier current flow and to reduce impedance of the connections. Higher current connections (not applicable to this design) need tracks with calculated widths.

Ground is usually the most important connection in a PCB layout. While it is difficult to describe all the rules, as they tend to interact and vary somewhat from design to design, the basic idea is to achieve a low impedance ground and minimize undesirable interaction between circuitry that can result from excessive current flow through the ground connections - especially under fault conditions.



The routes that were made visible using the 'Netlist-Show-All' command sequence are a guide to completing the PCB layout. By using the autorouter to only 'Process Pre-routes' instead of routing, we are able to eliminate the nets that have been placed as tracks already. As a result only those nets still to be placed are visible



When the board is complete, no more routes are visible. At this stage a Design Rule Check operation can be performed, this will test the PCB layout against the netlist and a set of rules previously specified. Any violations encountered will be reported (e.g. in a .DRC log file).

In a larger PCB design, there might be several iterations between schematic and PCB. For instance, it might be more convenient for layout to swop different gates (or SIP resistor) in a package, rather than use the one specified in the circuit. In the example CAD package shown, it is necessary to go back into the schematic package, edit the schematic, regenerate the netlist (using POST), and then re-import the netlist into the PCB CAD package. While this sounds arduous, it is quite easy to have all 3 programs running simultaneously under windows, and simply 'Alt-Tab' between them, so the modification is accomplished reasonably easily. Other CAD packages have 'back annotation' that accomplishes this task automatically.

Once the PCB has been completed and checked, a Bill-Of-Materials can be generated from it. In the case of the example CAD package, this is done by using the BOM.EXE utility.

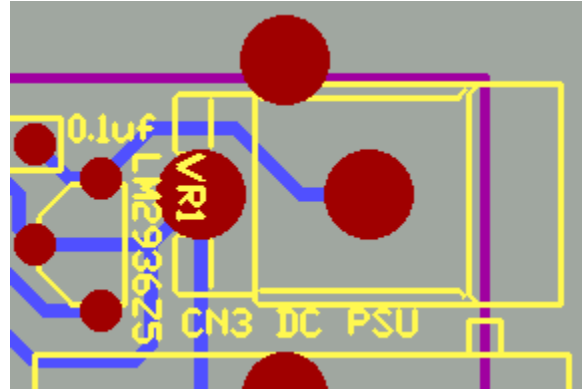
 232CNV.PCB 21:14 11-12-1998 Bill of Material Detail Page : 1

DESIGNATOR	COMMENT	PATTERN
A1	(C)1999	AIRBORN
C1	2.2UF	RB.1/.2
C2	2.2UF	RB.1/.2
C3	2.2UF	RB.1/.2
C4	2.2UF	RB.1/.2
C5	22UF	RB.1/.2
C6	0.1UF	RAD0.1
C7	0.1UF	RAD0.1
CN1	RS232	DB9RA/F
CN2	RS485	RJ12
CN3	DC PSU	DCSKT2.5
IC1	ICL232	DIP16
IC2	75176	DIP8
J1	TERMINATE	SIP2
J2	RI	SIP3
L1	232-RX	LED5
L2	485-RX	LED5
PCBLAM#1	AB9812222	PCLAMINATE
R1	120E	AXIAL0.4
SIP1	2.2K	SIP8
SIP2	2.2K	SIP8
TR1	BC557	TO-92A
VR1	LM2936Z5	TO-92A

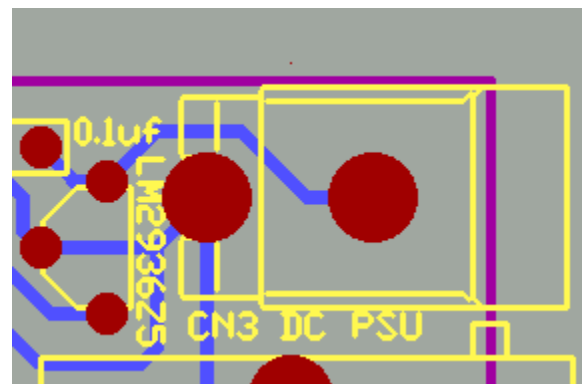
Total Components = 23

You can see that our PCLAMINATE component has carried through from the original schematic, through the Printed Circuit Board, into the Bill-Of-Materials. By using a unique number for each issue of PCB, we can tie all three items together with this component. Our copyright/logo has also appeared in all three files, using the same method.

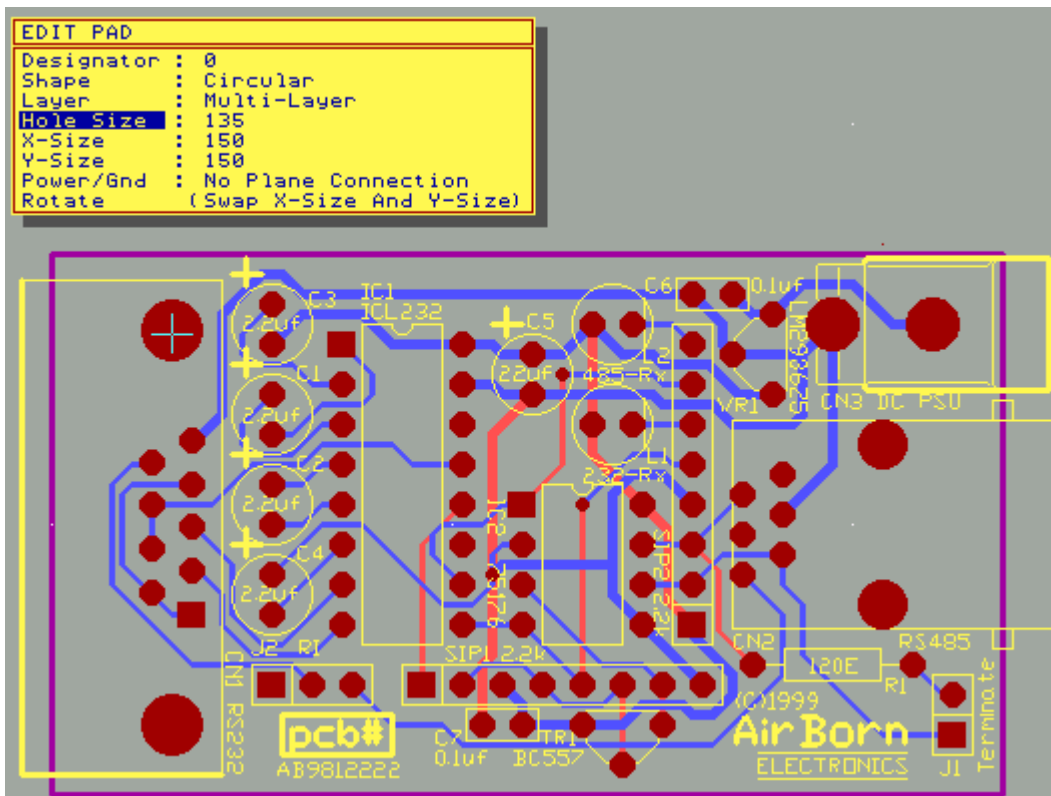
Our PCB still requires a little bit of cleaning up, as there is a pad overlapping the side of the PCB. While silkscreen legend can overlap the side of a PCB - it doesn't usually cause any problems - holes over the edge of the PCB will usually interfere with the routing or V-Grooving of the board.



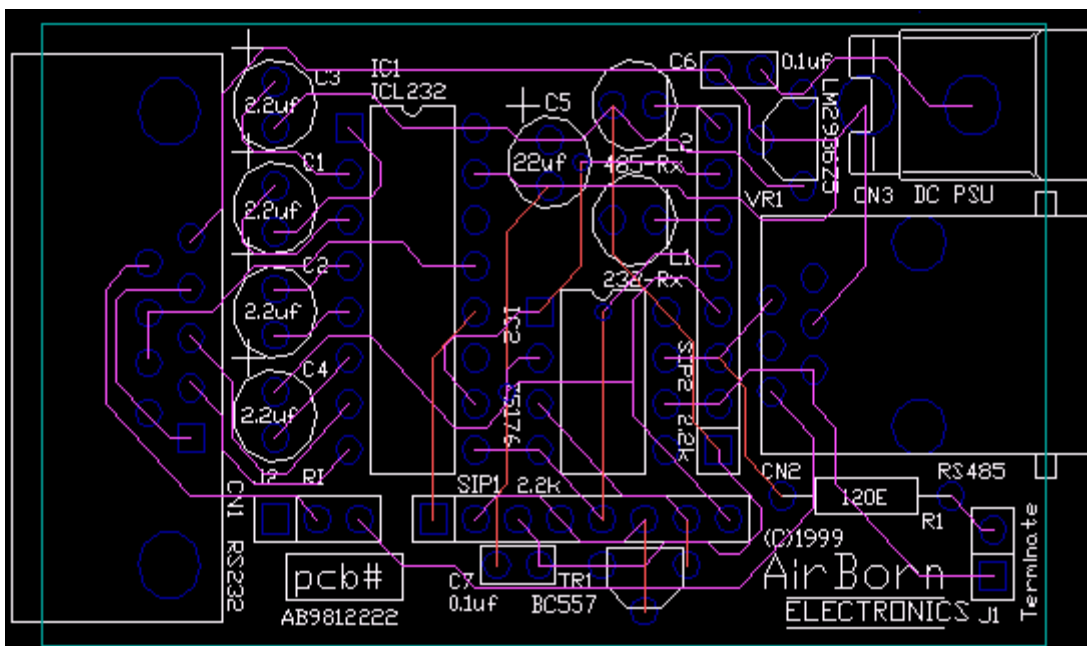
So we should fix this...



Next we check hole sizes of Pads on the PCB. The hole size required varies with the particular type of component. Incorrect hole size is a major source of error in producing PCBs.



If we have a manufacturer producing an enclosure for our electronic device, we can export an Autocad format mechanical file using the PCBTODXF.EXE utility.



Refer to our documentation page to see the total package that we put together for the Printed circuit board. The SCHPLOT.EXE utility is used to print the schematic diagram. The PLOT.EXE utility is used, with appropriate options selected, to print the checkplot (actual size and or enlarged as required) and the drilling diagram. As described above we use the BOM.EXE utility to print the parts list from the PCB itself, although we massage the result slightly using a program of our own called BOM2PART.EXE, to reorder the parts list by item (this was not necessary for our example PCB). We add an ECO notes sheet, a parts list key, and a

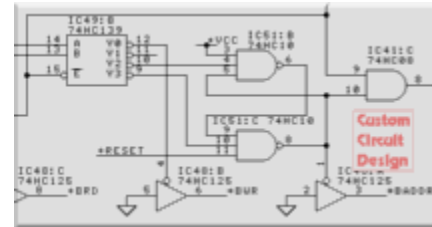
front cover, and then the documentation file for the Printed Circuit Board is complete.

Most CAD packages finish the job by saving the file as either native format, or preferably Gerber, to be sent to the PCB manufacturer to have the boards produced.

We also support the Autotrax/Easytrax CAD package by providing [Updated video drivers](#).

Click the *Continue* arrow below for "PCB Etching," or go on to:

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